

Rollback Overhead Reduction in Speculative Brain Simulators

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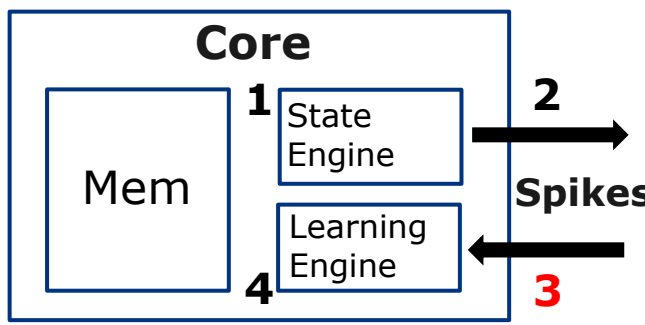
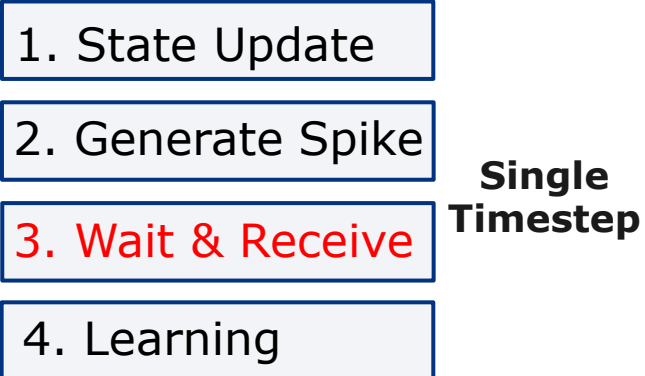
1 Background & Motivation

Brain Simulation

- Spiking neural networks (SNNs) model the brain as neurons exchanging discrete **spikes** over synapses
- Hardware-based **multicore** simulators are used for large-scale simulations

Bottleneck of Time-precise Simulation:

- Synchronization Barrier :
Early-finishing cores stall at global barrier, waiting all cores to finish computation & all spikes to reach destination cores
- Severe Resource Underutilization:
As the system scales, **~79%** of total cycles are wasted

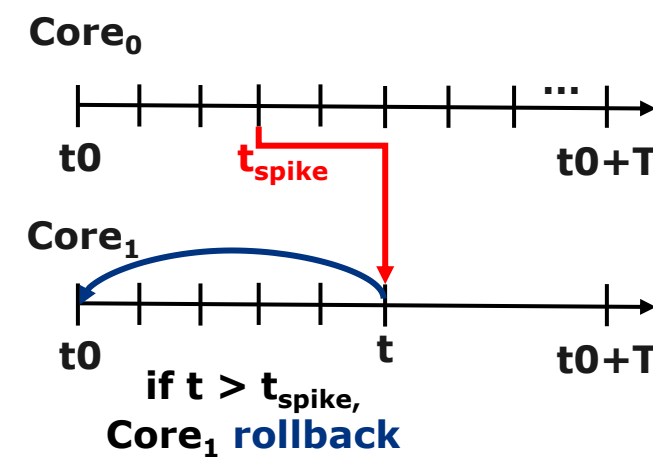
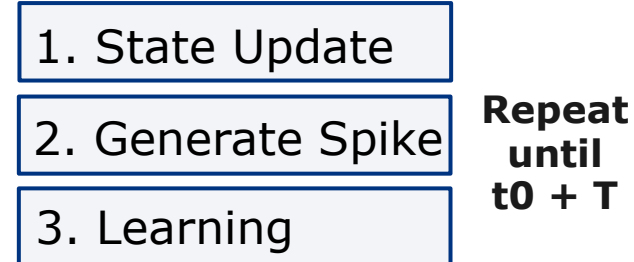


2 Baseline: Speculative brain simulator

NeuroSync¹ (HPCA'22)

Speculation mechanism

- Each core speculatively runs without synchronization within the global synchronization boundary **T**
- Rollback & Recovery :
when a core receives a **belated spike** – a remote spike that arrives after the core has already passed the target timestep – it **rolls back** its states to a clean checkpoint and **recovers** the affected computations

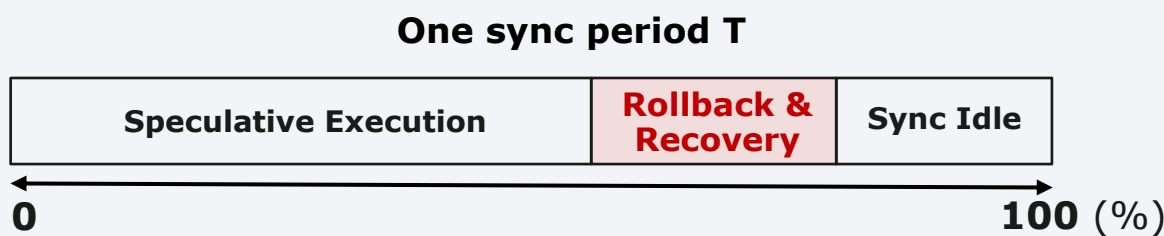


3 Key Observation

Hypothesis: rollback leads to unnecessary wasted latency – predicting or avoiding it leads to speedup

Measured: rollback = 10.4% of all cycles at the optimal $T = 64$

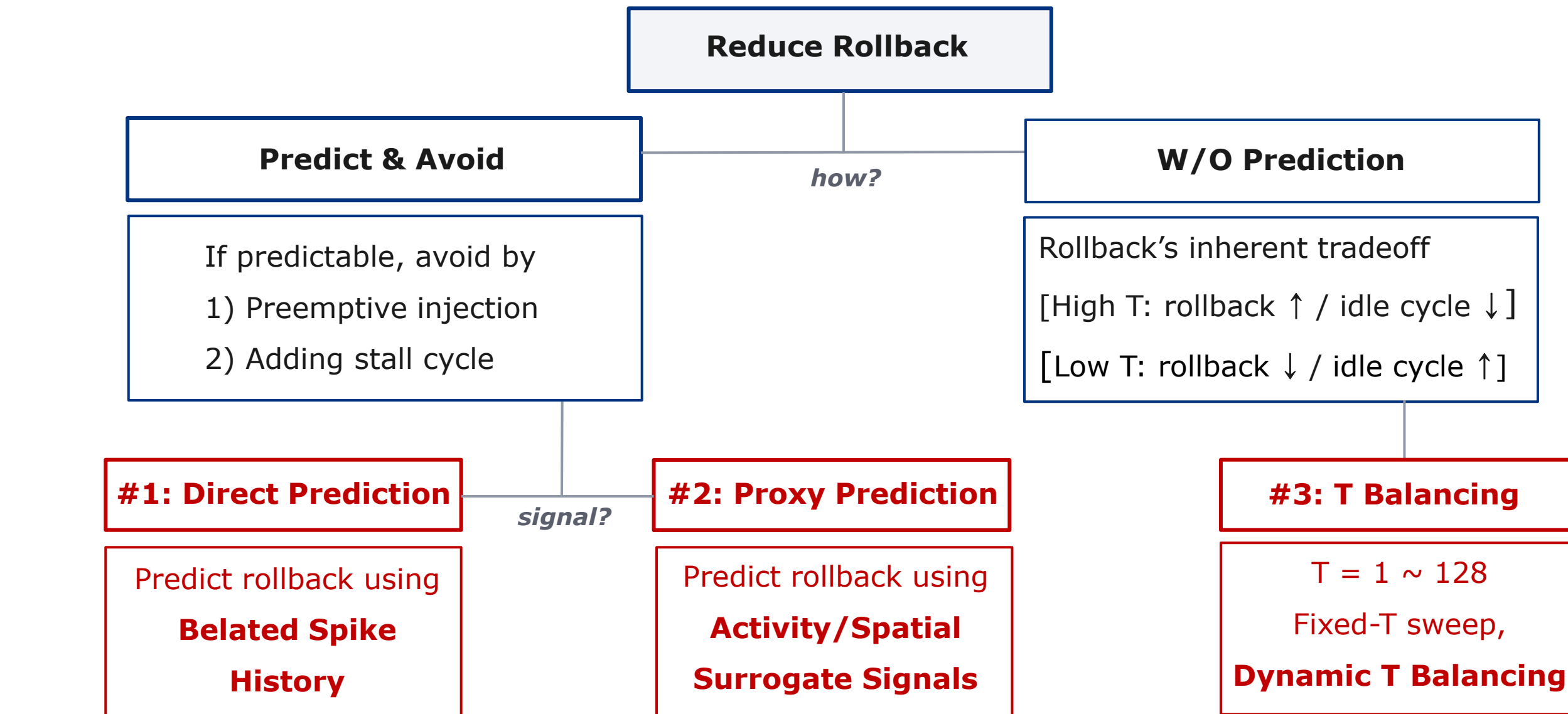
→ rising to **18.1%** at $T = 128$



Research question

Can rollback be **predicted**, or **balanced away**, to convert latency loss into speedup?

4 Three Key Ideas to Reduce Rollback



5 Idea #1: Belated-Spike Predictor

Perceptron Predictor

- What to predict:
Does core C at timestep t receives belated spike? (± 1)

Input Features

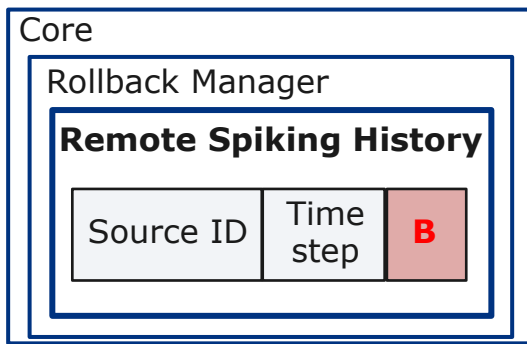
a) Local History

Simply add **"belated"** bit to existing hardware (Remote Spiking History) per core

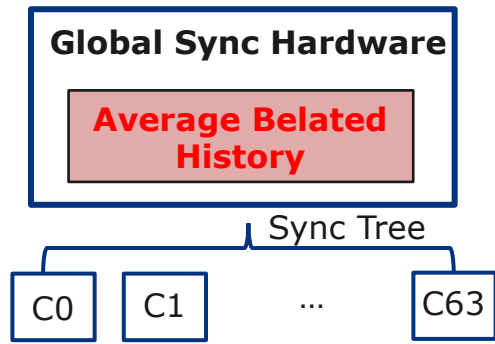
b) Global History

Add average belated history to global sync hardware

- Result : Predictor **AUC 0.53~0.58** (for $T = 64, 128$)



Per core Local History



Global History

6 Idea #2: Proxy Predictor

Intuition:

more spikes & longer **distances** → **Belated spikes**

Verifying the link:

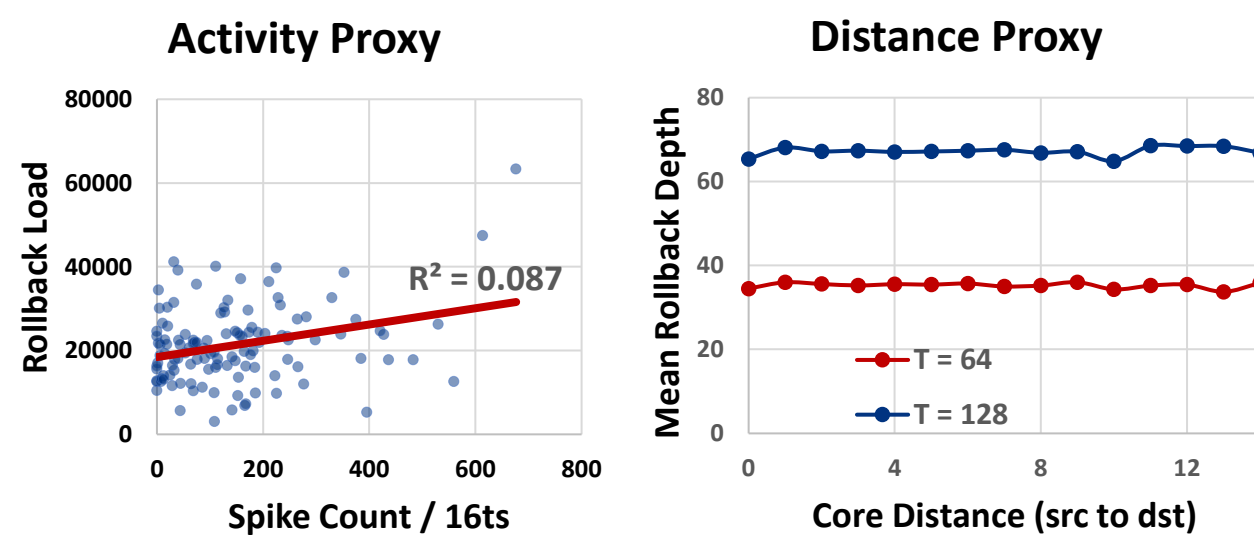
1) Activity : $R^2 = 0.09$

Network Activity (# of total spikes) ↔ Rollback Load

2) Spatial: $\text{corr} \approx 0$

inter-core spike transfer distance ↔ Rollback depth

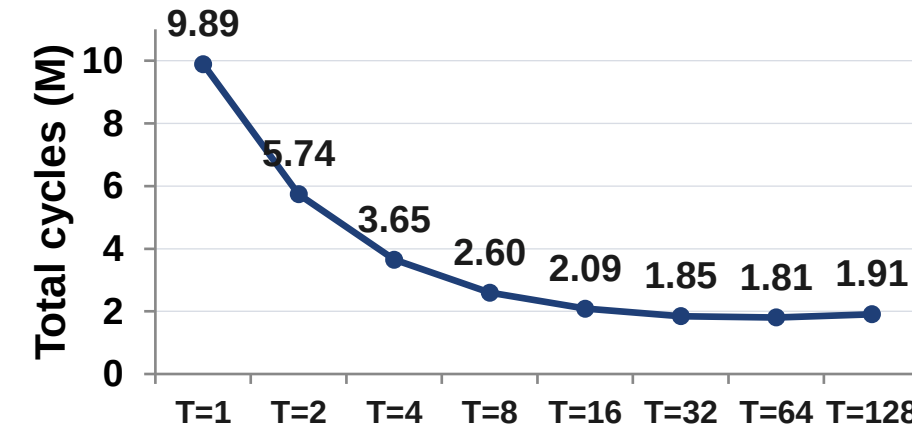
Link is very weak, proxy prediction is **unusable**



Rollback Load is calculated as $\sum(\text{rollback depth})$ for fixed timesteps

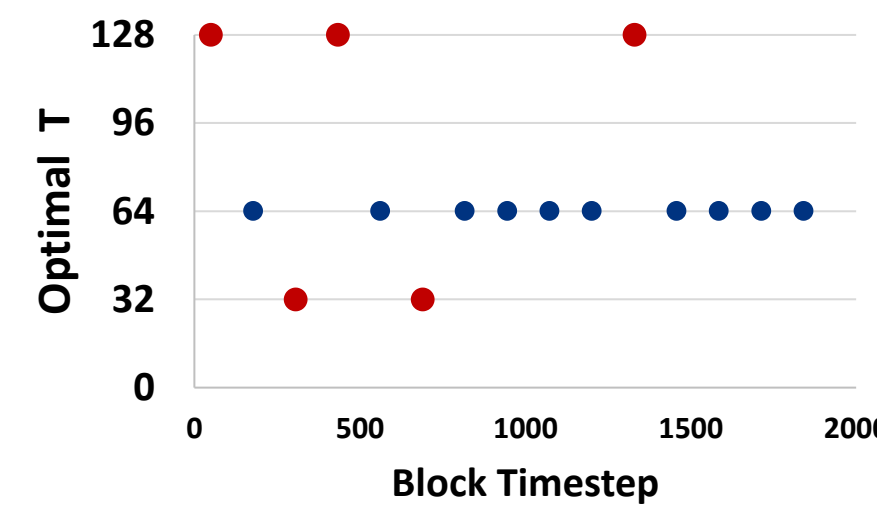
7 Idea #3: Dynamic T Balancing

Fixed-T sweep shallow U-curve:



Optimum at **T=64** with **1.81M** cycles

Oracle T Balancer:



Per 128 timestep block, zero T switching cost

Achieved **1.77M** cycles, **2.0% speedup only**

8 Evaluation Setup

Hardware:

64 cores (4 chips x 16) on 8x8 NoC mesh

Workload:

Brunel SI (sustained, inhibition-stabilized)
4100 neurons (E:3200 / I: 800 / P: 100)
1971 timestep simulation
(=50 warmup + 128 x 15 measurement)

Mapping:

Metis partition, ~64 neurons / core

Methodology:

Modified the open-source NeuroSync¹ simulator
Repository: github.com/SNU-HPCS/NeuroSync

[1] H. Lee et al., "NeuroSync: A Scalable and Accurate Brain Simulator Using Safe and Efficient Speculation," 2022 IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2022, pp. 633-647.

9 Analysis

Why did rollback prediction failed?

Micro-level non-linearity:

Even without considering belated phenomenon, individual spike firing rhythms are highly chaotic
Macro-level workload patterns are meaningless

Root Cause Identification:

Real-time NoC router contention was not considered.

Approaches without accurate prediction

~7.5 neurons per core receive belated spike:

64 neurons share a timestep
Conservative stalling is structurally inefficient

Negligible gains:

Hardware overhead required for T balancing outweighs the <2% benefit

Open: the energy trade-off is unquantified; less-homeostatic workloads are untested.

10 Conclusion & Future Work

Conclusion

- Perceptron Predictor achieved 0.53~0.58 AUC.
- Spike & NoC proxy has no correlation with rollback.
- Dynamic T balancing oracle achieved 2% speedup
- All three rollback reducing approaches are not effective.

Future Work

- Rather than predicting/reducing rollback itself, should focus on designing hardware to cut rollback overhead

Acknowledgement

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